

MOHAMMAD NAVED

Founder and Lead AI Engineer at Codtrex AI | Full-Stack Product Architect

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PROFILE SUMMARY

Founder and Lead AI Engineer with a track record of shipping production-grade full-stack products including a Google-certified IoT Smart Home ecosystem, a B2B cloud pharmacy SaaS, and an AI engineering studio. Founded Codtrex AI, a boutique studio delivering custom LLM integration, cloud systems, and IoT solutions for clients. Deep expertise across the full stack: embedded firmware (ESP32, Arduino), cloud infrastructure (AWS IoT Core, Docker, App Runner), modern web frontends (React 19, Next.js 15/16, TypeScript), and AI integration (Gemini API, multi-LLM orchestration). Actively seeking senior engineering roles and consulting engagements where ownership, zero-to-one velocity, and full-stack product thinking are valued.

CORE EXPERTISE

- **AI and LLM Engineering** - production Gemini API integration, multi-LLM orchestration, agentic workflows, prompt architecture, RAG system design
- **Full-Stack Product Development** - React 19, Next.js 15/16, TypeScript, Tailwind CSS v4, Vite, Supabase (PostgreSQL), FastAPI (Python)
- **IoT and Embedded Systems** - ESP32 firmware (Arduino/C++), MQTT protocol, AWS IoT Core, mTLS device authentication, OTA firmware updates
- **Cloud and DevOps** - AWS (IoT Core, App Runner, ECR, S3), Docker, Vercel, GitHub Actions CI/CD
- **Mobile Development** - React Native (Expo), Capacitor Android for Google Play
- **Studio and Client Delivery** - end-to-end product ownership from spec to deployment; Google Smart Home certification process

EXPERIENCE

Codtrex AI - Boutique AI Engineering Studio 2025 - Present

Founder and Lead Engineer | [codtrex.vercel.app](#) | React 19, Vite, Tailwind CSS v4, Supabase, Gemini API, Vercel

- Founded and operate a boutique AI engineering studio offering end-to-end LLM integration, full-stack web products, IoT systems, and technical consultation to clients
- Built and shipped the studio public website with an AI chat assistant, service tiers, and client case studies at [codtrex.vercel.app](#)
- Designed multi-LLM product architectures and client-facing AI solutions across web, mobile, and embedded platforms

4Layers - Google Certified Smart Home IoT Ecosystem 2025 - Present

Founder and Systems Engineer | ESP32, AWS IoT Core, MQTT, mTLS, React Native, FastAPI, Docker, AWS App Runner

- Achieved official Google Smart Home certification - passed 71 out of 75 tests in the Google Home Test Suite; devices listed and functional in the Google Home ecosystem
- Architected a 3-source real-time state synchronization system (ESP32 hardware to AWS IoT Core MQTT to Google Home) ensuring zero state divergence across physical buttons, app controls, and voice commands
- Engineered QA Hold Mode to prevent Google Home from reverting local hardware states during certification test execution - a critical blocking issue discovered and resolved independently
- Built OTA 2.0 with version pinning and rollback safety enabling firmware updates across deployed devices without physical access
- Designed custom ESP32 hardware nodes: 4-channel relay switch boards and Triac-based multi-speed fan regulator; mTLS certificates provisioned per device via AWS IoT Core

MediStock - Cloud Pharmacy Management SaaS 2025 - Present

Founder and Product Architect | React 18, Supabase, PostgreSQL, Capacitor Android, Vercel | [medistock-pharma.vercel.app](#)

- Built and shipped a full B2B pharmacy SaaS including inventory management, expiry tracking, billing, and supplier management for pharmacy businesses
- Deployed cross-platform as a live web app and Android APK via Capacitor targeting Google Play submission
- Designed multi-tenant Supabase (PostgreSQL) schema with Row Level Security for pharmacy data isolation

PROJECTS

UsedTech Market - C2C Electronics Marketplace 2025

Product Architect | Next.js 15/16, React 19, TypeScript, Supabase, Tailwind CSS v4

- Architected a consumer-to-consumer marketplace for buying and selling used electronics with real-time listings, search filters, and seller dashboards
- Built on Next.js 15/16 App Router with SSR for SEO-optimized product pages and Supabase real-time subscriptions for live listing updates

AutoApply AI - Automated Job Application System 2025

Self-Initiated | Python, Gemini AI, Multi-LLM Orchestration

- Designed a multi-step prompt chain that reads job descriptions, extracts requirements, and generates tailored resumes and cover letters via structured LLM calls
- Engineered retry and fallback prompt logic to handle malformed or incomplete model outputs in production use

Self-Initiated | React, TypeScript, Recharts, Gemini AI, DuckDB, SQL

- Built an AI analytics platform with a natural-language-to-SQL chat interface and an ML Lab module for model output explanation and insight surfacing
- Designed prompt templates for consistent LLM output across varying data schemas and edge cases

TECHNICAL SKILLS

AI and LLM: Gemini API, Multi-LLM Orchestration, Prompt Engineering, Agentic Workflows, RAG Systems

Frontend: React 19, Next.js 15/16, TypeScript, Tailwind CSS v4, Vite, Framer Motion

Backend: FastAPI (Python), Node.js, REST APIs, Supabase, PostgreSQL, SQL

IoT and Embedded: ESP32 (Arduino/C++), MQTT, AWS IoT Core, mTLS, OTA Firmware Updates, Triac control

Cloud and DevOps: AWS IoT Core, AWS App Runner, AWS ECR, AWS S3, Docker, Vercel, GitHub Actions

Mobile: React Native (Expo), Capacitor Android

Languages: TypeScript, Python, C++ (Arduino), SQL, JavaScript

EDUCATION

B.Tech, Computer Science - RGPV University, Indore

2025

CERTIFICATIONS AND ACHIEVEMENTS

- Google Smart Home Certification - 4Layers IoT Ecosystem (71 out of 75 tests passed, Google Home Test Suite)
- Generative AI with Gemini - Prompt Engineering and LLM Application Development
- Python for Data Analysis and Visualization - Self-directed
- SQL for Data Analytics - Relational Database Workflows
- Power BI - Interactive Dashboard Design and Reporting